

Zhaofeng (Philip) Wang

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Education

Northwestern University, PhD in ECE, *Advised by Prof. Qi Zhu*

Sept. 2026 - Present

Northwestern University, BA in Mathematics and Psychology

Sept. 2021 - June 2025

- GPA: 3.9/4.0

- **Relevant Coursework:** Probability and Stochastic Processes, Graduate Seminar in Algorithm, Graduate Seminar in Statistical Network Analysis, Probabilistic Graphical Models, Game Theory and Networked Systems

Research Interests

Reinforcement learning, formal methods, and control for safe autonomous systems. I integrate formal verification and control-theoretic tools with learning-based methods to develop safe, interpretable, and scalable embodied agents.

Research Experience

IDEAS Lab, Northwestern University, Evanston, IL

Jan. 2024 - Present

Advised by Prof. Qi Zhu

- Lead a team of five master's and undergraduate students developing **ManiGuard**, a robotic manipulation safety benchmark with 200 tasks, 1,000 evaluation scenarios, and 8,000 safety-annotated demonstrations; co-developed specification-grounded runtime monitoring to evaluate safety independently of task success.
- Co-developed **SENTINEL (NeurIPS 2026)**, using temporal logic to evaluate foundation model-based agents across semantic interpretation, planning, and execution, with counterexample feedback for safety improvement.
- Developed transition-aware reward shaping for **Model-Enhanced IRL (L4DC 2026)**, establishing theoretical guarantees and improving sample efficiency and performance in stochastic MuJoCo and Atari environments.
- Co-developed **belief-based offline RL (ICLR 2026)** using transformer-based belief prediction for delay-robust deployment from delay-free data; developed inverse RL methods for delayed demonstrations.

Center for Deep Learning, Northwestern University, Evanston, IL

Aug. 2025 - June 2026

Research Assistant, *Advised by Prof. Zhaoran Wang*

- Proposed a hierarchical latent-token reasoning framework for transformers and empirically validated it on a synthetic dataset with natural language evidence.
- Created a synthetic two-level generative dataset with explicit latent ground truth for interpretable circuit tracing using a transcoder-based approach.
- Analyzed circuit formation and token-wise behaviors during training; identified reasoning patterns systematically through circuit clustering.

Causal Inference Lab, Northwestern University, Evanston, IL

Mar. 2024 - Aug. 2024

Undergraduate Researcher, *Advised by Prof. Zach Wood-Doughty*

- Augmented MIMIC-III benchmarks for medical LLMs in collaboration with physicians to prevent shortcut learning when evaluating clinical reasoning across NLP tasks.
- Fine-tuned transformer models with LoRA, achieving state-of-the-art results on existing benchmarks and demonstrating a more than 30% performance decrease on the augmented dataset.
- Conducted statistical analysis using scikit-learn to assess consistency across physician annotations and quantify discrepancies between human and LLM annotations.

Teaching Experience

Peer Mentor, CS_474 Probabilistic Graphical Models, Northwestern University

Fall 2024

Academic Advisor, Intensive Law & Trial, Stanford Law School

Summer 2022, 2023

Publications (* indicates equal contribution)

ManiGuard: A Benchmark and Data Suite for Specification-Grounded Safety Evaluation and Improvement of Robotic Manipulation.

Yiyan Peng *, **Philip Wang** *, Simon Sinong Zhan *, Yiqi Lyu, Zhenyang Ni, Jixin Yan, Fiorelli Wong, Ruochen Jiao, Hang Yin, Xinyu Cao, Huajie Shao, Manling Li, Ruohan Zhang, Qi Zhu.

arXiv 2026 | [Paper](#) | [Project](#) | [Code](#)

SENTINEL: A Multi-Level Formal Framework for Safety Evaluation of Foundation Model-based Embodied Agents.

Simon Sinong Zhan *, **Philip Wang** *, Justin Liu *, Yiyan Peng *, Yiqi Lyu, Zinan Wang, Qineng Wang, Zhian Ruan, Xiangyu Shi, Xinyu Cao, Frank Yang, Zhenyang Ni, Kangrui Wang, Ruohan Zhang, Huajie Shao, Manling Li, Qi Zhu.

NeurIPS 2026 | [Paper](#) | [Project](#) | [Code](#)

Belief-Based Offline Reinforcement Learning for Delay-Robust Policy Optimization.

Simon Sinong Zhan *, Qingyuan Wu *, **Philip Wang**, Frank Yang, Xiangyu Shi, Chao Huang, Qi Zhu.

ICLR 2026 | [Paper](#)

Enhancing Inverse Reinforcement Learning through Encoding Dynamic Information in Reward Shaping.

Simon Sinong Zhan *, **Philip Wang** *, Qingyuan Wu *, Ruochen Jiao, Yixuan Wang, Chao Huang, Qi Zhu.

L4DC 2026 | [Paper](#)

Inverse Delayed Reinforcement Learning.

Simon Sinong Zhan *, Qingyuan Wu *, Zhian Ruan, Frank Yang, **Philip Wang**, Yixuan Wang, Ruochen Jiao, Chao Huang, Qi Zhu.

arXiv 2024 | [Paper](#)

Towards Rigorously Evaluating Clinical Reasoning in LLMs.

Christopher Wong, Divy Kumar*, Amiin Muse*, **Philip Wang***, Finn Wintz*, Zach Wood-Doughty.

Leadership Experience

Treasurer and Competing Attorney & Witness, Northwestern Mock Trial Team Sept. 2021 - June 2025

- Managed a budget of over \$50,000; revised the budget structure, reducing student membership fees by 20%.
- Placed 8th out of approximately 700 teams at the American Mock Trial Association National Championship Tournament.
- Instructed 120 high school students across six ten-day summer sessions at Stanford Law School in mock trial, legal research, public speaking, and argument drafting.

Residential Assistant, Northwestern Residential Services Aug. 2022 - June 2025

- Advised 150 residents over three years, supporting community, individual growth, diversity, and well-being.
- Connected residents with campus organizations and resources in crises and everyday situations.
- Managed community funds and organized biweekly community events.

Awards

Northwestern Computer Science Department Summer Research Grant	2024
Sponsorship to compete in United States Bridge Championships	2024, 2025
Award for Excellence in Mathematics by a First-Year Student	2022
Selected for US U21 National Bridge Team twice	2019, 2022

Technologies

Languages: Python, C++, Java, Lean

Packages and Frameworks: PyTorch, scikit-learn, OpenCV, OpenAI Gymnasium, Stable Baselines, AI2Thor, BEHAVIOR-1k